

Java Stream API Cheat Sheet

1. Find First Non-Repeated Character

```
Optional<Character> firstNonRepeated = input.chars()
    .mapToObj(c -> (char) c)
    .collect(Collectors.groupingBy(c -> c, LinkedHashMap::new, Collectors.counting()))
    .entrySet().stream()
    .filter(e -> e.getValue() == 1)
    .map(Map.Entry::getKey)
    .findFirst();
```

2. Find Duplicates in a List

```
Set<String> duplicates = list.stream()
    .collect(Collectors.groupingBy(Function.identity(), Collectors.counting()))
    .entrySet().stream()
    .filter(e -> e.getValue() > 1)
    .map(Map.Entry::getKey)
    .collect(Collectors.toSet());
```

3. Group Strings by Length

```
Map<Integer, List<String>> grouped = list.stream()
    .collect(Collectors.groupingBy(String::length));
```

4. Count Word Occurrences

```
Map<String, Long> wordCounts = list.stream()
    .collect(Collectors.groupingBy(Function.identity(), Collectors.counting()));
```

5. Sort by Length then Alphabetically

```
List<String> sorted = list.stream()
    .sorted(Comparator.comparingInt(String::length)
    .thenComparing(Comparator.naturalOrder()))
    .collect(Collectors.toList());
```

6. Find Max Element

```
int max = list.stream()
    .max(Integer::compareTo)
    .orElseThrow();
```

7. Join Strings with Comma

```
String result = list.stream()
    .collect(Collectors.joining(", "));
```

8. Flatten List of Lists

```
List<String> flat = nestedList.stream()
    .flatMap(List::stream)
    .collect(Collectors.toList());
```

9. Find Second Highest Number

```
Optional<Integer> secondHighest = list.stream()  
    .distinct()  
    .sorted(Comparator.reverseOrder())  
    .skip(1)  
    .findFirst();
```

10. Partition Even and Odd

```
Map<Boolean, List<Integer>> partitioned = list.stream()  
    .collect(Collectors.partitioningBy(n -> n % 2 == 0));
```

11. Remove Nulls

```
List<String> cleaned = list.stream()  
    .filter(Objects::nonNull)  
    .collect(Collectors.toList());
```